

Permutation and Combinations:

Permutation:

Any arrangement of a set of n objects in a given order is called Permutation of Object. Any arrangement of any $r \leq n$ of these objects in a given order is called an r permutation or a permutation of n object taken r at a time.

It is denoted by $P(n, r)$

$$P(n, r) = \frac{n!}{(n-r)!}$$

Theorem: Prove that the number of permutations of n things taken all at a time is $n!$.

Proof: We know that

$${}_nP_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = \frac{n!}{1} = n!$$

Example: $4 \times {}_nP_3 = {}_{n+1}P_3$

Solution: $4 \times \frac{n!}{(n-3)!} = \frac{(n+1)!}{(n+1-3)!}$

$$\frac{4 \times n!}{(n-3)!} = \frac{(n+1) \times n!}{(n-2)(n-3)!}$$

$$4(n-2) = (n+1)$$

$$4n - 8 = n + 1$$

$$3n = 9$$

$$n = 3.$$

Permutation with Restrictions:

The number of permutations of n different objects taken r at a time in which p particular objects do not occur is

$${}_n P_{r-p}$$

The number of permutations of n different objects taken r at a time in which p particular objects are present is

$${}_n P_{r-p} \times {}_p P_p$$

Example: How many 6-digit numbers can be formed by using the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 if every number is to start with '30' with no digit repeated?

Solution: All the numbers begin with '30'. So, we have to choose 4-digits from the remaining 7-digits.

∴ Total number of numbers that begins with '30' is

$${}_{7P_4} = \frac{7!}{(7-4)!} = \frac{7 \times 6 \times 5 \times 4 \times 3!}{3!} = 840.$$

Permutations with Repeated Objects:

Theorem: Prove that the number of different permutations of n distinct objects taken at a time when every object is allowed to repeat any number of times is given by n^r .

Proof: Assume that with n objects we have to fill r place when repetition of the object is allowed.

Therefore, the number of ways of filling the first place is $= n$

The number of ways of filling the second place $= n$

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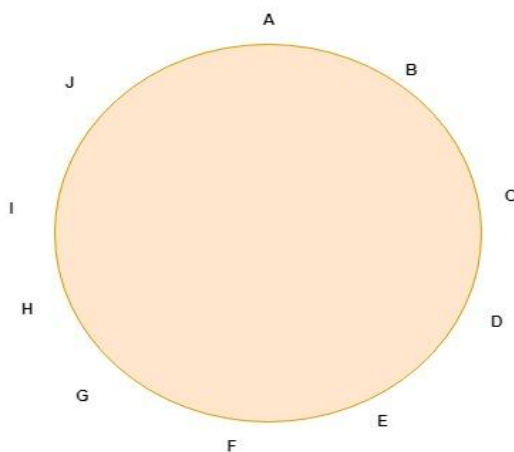
The number of ways of filling the r th place $= n$

Thus, the total number of ways of filling r places with n elements is

$$= n \cdot n \cdot n \dots \dots \dots r \text{ times} = n^r.$$

Circular Permutations:

A permutation which is done around a circle is called Circular Permutation.



Example: In how many ways can get these letters a, b, c, d, e, f, g, h, i, j arranged in a circle?

Solution: $(10 - 1) = 9! = 362880$

Theorem: Prove that the number of circular permutations of n different objects is $(n-1)!$

Proof: Let us consider that K be the number of permutations required.

For each such circular permutations of K , there are n corresponding linear permutations. As shown earlier, we start from every object of n object in the circular permutations. Thus, for K circular permutations, we have $K \dots n$ linear permutations.

Therefore, $K \cdot n = n!$ or $K = \frac{n!}{n}$

$$K = \frac{n \times (n-1)!}{n}$$

$$K = (n-1)!$$

Hence Proved.

Combination:

A Combination is a selection of some or all, objects from a set of given objects, where the order of the objects does not matter. The number of combinations of n objects, taken r at a time represented by n_{C_r} or $C(n, r)$.

$$n_{C_r} = \frac{n!}{r!(n-r)!}$$

Proof: The number of permutations of n different things, taken r at a time is given by

$$n_{P_r} = \frac{n!}{(n-r)!}$$

As there is no matter about the order of arrangement of the objects, therefore, to every combination of r things, there are $r!$ arrangements i.e.,

$$n_{P_r} = r! \cdot n_{C_r} \quad \text{or} \quad n_{C_r} = \frac{n_{P_r}}{r!} = \frac{n!}{(n-r)!r!}, n \geq r$$

Thus,

$$n_{C_r} = \frac{n!}{r!(n-r)!}$$

Example: A farmer purchased 3 cows, 2 pigs, and 4 hens from a man who has 6 cows, 5 pigs, and 8 hens. Find the number m of choices that the farmer has.

The farmer can choose the cows in $C(6, 3)$ ways, the pigs in $C(5, 2)$ ways, and the hens in $C(8, 4)$ ways. Thus the number m of choices follows:

$$m = \binom{6}{3} \binom{5}{2} \binom{8}{4} = \frac{6 \cdot 5 \cdot 4}{3 \cdot 2 \cdot 1} \times \frac{5 \cdot 4}{4} \times \frac{8 \cdot 7 \cdot 6 \cdot 5}{4 \cdot 3 \cdot 2 \cdot 1} = 20 \times 10 \times 70 = 14000$$